
Assistant Professor
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EMPLOYMENT

03/2026 –	Boya Young Scholar, Assistant Professor Department of Atmospheric and Oceanic Sciences, School of Physics Peking University
07/2025 – 02/2026	Postdoctoral Associate Department of Earth, Atmospheric & Planetary Sciences Massachusetts Institute of Technology Host: Prof. Abigail Bodner
06/2023 – 06/2025	Houghton Postdoctoral Fellow Department of Earth, Atmospheric & Planetary Sciences Massachusetts Institute of Technology Host: Prof. Raffaele Ferrari

EDUCATION

09/2018 – 06/2023	Department of Atmospheric and Oceanic Sciences, University of California, Los Angeles <i>Ph.D. in Atmospheric and Oceanic Sciences (Jun 2023)</i> <i>M.S. in Atmospheric and Oceanic Sciences (Dec 2020)</i> Advisor: Prof. Andrew L. Stewart
09/2014 – 07/2018	School of Physics, Peking University <i>B.S. in Astronomy (with honors)</i>

PUBLICATIONS

* = Corresponding Author, † = Student Advisees

Manuscripts in preparation:

1. **Y. Si*** and R. Ferrari. Parametric Instability of a Tidal Flow in a Submarine Canyon. *In preparation for the Journal of Physical Oceanography*. (Preprint available upon request)

2. Z. Zhang* et al. (including **Y. Si**). Water-Mass Exchanges between the Antarctic Continental Shelves and Open Ocean. *In preparation for Nature Reviews Earth & Environment*.

Submitted:

3. **Y. Si**, L. Johnson, and A. Bodner*. Uncovering Ocean Mixed Layer Dynamics from the Sea Surface State. *Submitted*. [[Preprint](#)]
4. **Y. Si***, R. Ferrari, and G. Voet. Tidally Induced Turbulence in the Abyssal Ocean. *Submitted*. [[Preprint](#)]

Peer-reviewed:

5. M. K. Youngs*, A. L. Stewart, **Y. Si**, A. F. Thompson, and M. P. Schodlock. Antarctic ice-shelf basal melt shaped by competing feedbacks. *Nature Geoscience*, 2026. [[doi](#)]
6. S. Spungin*[†], **Y. Si**, A. L. Stewart, and C. J. Prend. Observed Seasonality of Mixed-Layer Eddies and Vertical Heat Transport over the Antarctic Continental Shelf. *Journal of Geophysical Research: Oceans*, 2025. [[doi](#), [code](#)] ([†]*Student co-advised by A.L. Stewart and Y. Si*)
7. X. Ruan*, **Y. Si**, and R. Ferrari. Diapycnal upwelling driven by tidally-induced mixing over steep topography. *Journal of Physical Oceanography*, 2025. [[doi](#), [pdf](#)]
8. **Y. Si***, A. L. Stewart, A. Silvano, and A. C. Naveira Garabato. Antarctic slope undercurrent and onshore heat transport driven by ice shelf melting. *Science Advances*, 2024. [[doi](#), [pdf](#), [code](#), [data](#)]
9. C. R. Schmidgall*[†], **Y. Si**, A. L. Stewart, A. F. Thompson, and A. McC. Hogg. Dynamical controls on bottom water transport and transformation across the Antarctic Circumpolar Current. *Journal of Physical Oceanography*, 2023. [[doi](#), [pdf](#)] ([†]*Student co-advised by A.L. Stewart and Y. Si*)
10. **Y. Si***, A. L. Stewart, and I. Eisenman. Heat transport across the Antarctic Slope Front controlled by cross-slope salinity gradients. *Science Advances*, 2023. [[doi](#), [pdf](#), [code](#), [data](#)]
11. A. Silvano*, P. Holland, K. Naughten, O. Dragomir, P. Dutrieux, A. Jenkins, **Y. Si**, A. L. Stewart, B. Pena-Molino, G. Janzing, T. Dotto, and A. C. Naveira Garabato. Baroclinic ocean response to climate forcing regulates decadal variability of ice-shelf melting in the Amundsen Sea. *Geophysical Research Letters*, 2022. [[doi](#)]
12. **Y. Si***, A. L. Stewart, and I. Eisenman. Coupled ocean/sea ice dynamics of the Antarctic Slope Current driven by topographic eddy suppression and sea ice momentum redistribution. *Journal of Physical Oceanography*, 2022. [[doi](#), [pdf](#), [code](#), [data](#)]
13. **Y. Si**, J. Yang*, and Y. Liu. Planetary climate under extremely high vertical diffusivity. *Astronomy & Astrophysics*, 2022. [[doi](#), [pdf](#), [code](#)]

Non-Peer Reviewed:

- Y. Si. Ice-Ocean Interactions in the Antarctic Slope Current. *Ph.D. Dissertation*, University of California, Los Angeles, 2023. [[online](#)]
- Y. Si. Tides, Oceans, and Climate on Exoplanets. *B.S. Thesis*, Peking University, 2018.

SELECTED AWARDS

11/2025	NSFC Excellent Young Scientists Fund (Overseas)
06/2023 – 06/2025	MIT Distinguished Postdoctoral Fellowship in Earth, Atmospheric & Planetary Sciences
01/2024	UCLA Jacob A. Bjerknes Award
10/2022 – 06/2023	UCLA Dissertation Year Fellowship
09/2018 – 08/2022	China Scholarship Council Fellowship
2015 – 2018	Beijing Outstanding Graduate Award (07/2018) Peking University: Outstanding Graduate Award (07/2018), May 4th Scholarship (10/2017), Outstanding Student Award (10/2017), Social Work Award (10/2015) School of Physics: Outstanding Bachelors Thesis (06/2018), Wei-Ming Scholarship (11/2017) The 5th Conference on Earth System Science: Outstanding Student Research (07/2018) Chinese Meteorological Society: Outstanding Oral Presentation Award (09/2017)

RESEARCH SUPPORT

2026 – 2029	Ice-Ocean Interactions in the Polar Oceans (PI, NSFC Excellent Young Scientists Fund)
2023 – 2026	Tidally Induced Turbulence over Sloping Topography (Project Lead, NCAR Large Allocation, 19.3 million CPU hours + 17,500 GPU hours + 70 TB storage)

MENTORSHIP

05/2026 –	Jinze Du, undergraduate researcher, School of Physics, Peking University Project: <i>Machine Learning-Based Parameterization of Ice Shelf Basal Melting</i>
04/2026 –	Bohua Wang, undergraduate researcher, School of Physics, Peking University Project: <i>Tide-Topography Interactions on Ice-Covered Moons</i>
03/2026 –	Qinxue Wang, undergraduate researcher, Yuanpei College, Peking University Project: <i>Submesoscale-Induced Phytoplankton Bloom</i>
03/2026 –	Yuxuan Yang, undergraduate researcher, School of Physics, Peking University Project: <i>Tidally Induced Melt in the Ice Shelf Basal Channels</i>
03/2022 – 06/2023	Sophia Spungin, undergraduate researcher (Co-advised by Prof. Andrew Stewart), UCLA Project: <i>Mixed-Layer Eddies over the Antarctic Continental Shelf</i>
01/2020 – 03/2022	Carlyn Schmidgall, undergraduate researcher (Co-advised by Prof. Andrew Stewart), UCLA Project: <i>Transport and transformation of AABW across the ACC</i>

TEACHING

- Instructor: Intriguing Ocean Phenomena and Their Physical Principles. Fall 2026 (*scheduled*) — Undergraduate
- Instructor: Geophysical Fluid Dynamics. Spring 2027 (*scheduled*) — Graduate
- Teaching Associate/Assistant: Physical Oceanography, UCLA AOS 103. Fall 2019, 2020, 2021.
- Teaching Assistant: Air Pollution, UCLA AOS 2. Spring 2019.

UNIVERSITY SERVICE

School & Department Services:

- Undergraduate Student Advising Sessions: Provided academic and career advising to undergraduate students in the School of Physics, Peking University since Apr 2026 (7 sessions in total) 北京大学物理学院综合指导课 (累积指导本科生7人次)
- Judge, CUPT Peking University On-campus Competition & 15th Undergraduate Mini-Project Showcase, School of Physics, Peking University, 16 May 2026 北京大学物理学院第15届本科生小型项目训练与成果展示 (CUPT校内赛) 评委
- “Meteorology Month” Outreach Event, Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University, Apr 22, 2026. 北京大学物理学院大气与海洋科学系“气象月”活动
- Undergraduate Open Day at the Department of Atmospheric and Oceanic Sciences (19 students), School of Physics, Peking University, Apr 19, 2026. 北京大学物理学院大气与海洋科学系本科生开放日
- *Introduction to the Department of Atmospheric and Oceanic Sciences* (~30 undergraduate students), School of Physics, Peking University, Mar 20, 2026. Delivered on behalf of Prof. Yang. 大气与海洋科学专业分流宣讲
- *Introduction to Physical Oceanography and Paleoclimatology*, Seminar for Undergraduate Students (~100 students), School of Physics, Peking University, Mar 12, 2026. 北京大学物理学院现代物理前沿讲座II
- *Introduction to Physical Oceanography*, Workshop for Undergraduate Research (73 students), Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University, Jan 12–15, 2026. 北京大学物理学院大气与海洋科学系本科生科研素养寒假学堂

PhD Thesis Committee Member:

- Dr. Jie Mei (Advisor: Prof. Yonggang Liu), Spring 2026

ACADEMIC SERVICE

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| Dec 9–13, 2024 | Co-convener for the AGU Annual Meeting Session: <i>Ice-Ocean interactions along Antarctic Continental Shelf</i> |
| Since 2022 | Reviewer for the <i>Journal of Physical Oceanography</i> , <i>Geophysical Research Letters</i> , <i>Journal of Geophysical Research: Oceans</i> , <i>Journal of Climate</i> , and <i>Earth's Future</i> . |

SELECTED ABSTRACTS (PRESENTER)

- *What Triggers Intense Turbulent Overtakes in the Abyssal Ocean? An Integrated Perspective from Observations, Simulations, and Theory*. Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University, Jan 24, 2025. (Seminar)
- *Ocean Heat Transport and Antarctic Ice Shelf Melt: Two Positive Feedback Mechanisms*, School of Oceanography, Shanghai Jiao Tong University, Jan 2, 2025. (Seminar)
- *Tidally Induced Turbulence over Sloping Topography*, AGU Annual Meeting, Washington, D.C., Dec 9-13, 2024. (Talk)
- *Tidally Induced Turbulence over Sloping Topography*, Ocean Mixing Gordon Research Conference, South Hadley, MA, Jun 9-14, 2024. (Poster)
- *Ocean Heat Transport and Antarctic Ice Shelf Melt: Two Positive Feedback Mechanisms*, Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University, Jan 3, 2024. (Seminar)
- *Antarctic slope undercurrent and onshore heat transport driven by ice shelf melting*, 2nd US Antarctic Science Meeting, Jun 20-23, 2023. (Talk)
- *Antarctic slope undercurrent and onshore heat transport driven by meltwater upwelling*, Climate Dynamics Group, Jun 6, 2023. (Seminar)
- *Interactions between Antarctic continental shelf/slope circulation and ice shelf melt*, Department of Atmospheric & Oceanic Sciences, UCLA, Apr 5, 2023. (Seminar)
- *Drivers of the West Antarctic Undercurrent and Heat Transport toward Ice Shelves*, AGU Fall Meeting, Chicago, IL, Dec 11-16, 2022. (Poster)
- *Toward Circumpolar Regional Modeling of “Hot Spots” in Onshore Ocean Heat Transport*, AGU Fall Meeting, Chicago, IL, Dec 11-16, 2022. (Poster)
- *Heat transport across the Antarctic Slope Front*, Scientific Committee on Antarctic Research (SCAR) Meeting, Aug 1-10, 2022. (Talk)
- *Heat transport across the Antarctic Slope Front controlled by cross-slope salinity gradients*, the 23rd Conference on Atmospheric and Oceanic Fluid Dynamics, Breckenridge, CO, Jun 13-17, 2022. (Talk)
- *Coastal freshening enhances eddy-driven heat transfer toward the Antarctic margins*, AGU Ocean Sciences Meeting, Feb 24-Mar 4, 2022. (Talk)
- *Coastal freshening enhances eddy-driven heat transfer toward the Antarctic margins*, Peking University-Fudan University AOS Symposium, Jan 14, 2022. (Seminar)
- *Heat transport across the Antarctic Slope Front controlled by cross-slope salinity gradients*, Alfred Wegener Institute, Helmholtz Centre for Polar and Marine Research, Nov 3, 2021. (Seminar)
- *Heat transport across the Antarctic Slope Front controlled by cross-slope salinity gradients*, California Geophysical Fluid Dynamics (CalGFD) Meeting, Sep 2-3, 2021. (Talk)

- *Heat transport across the Antarctic Slope Front controlled by cross-slope salinity gradients*, US Antarctic Science Meeting (US-SCAR), Jul 12-15, 2021. (Talk)
- *Coupled ocean/sea ice circulation in the Antarctic Slope Current due to topographic eddy suppression and sea ice momentum redistribution*, Climate Dynamics Group, Apr 10, 2021. (Seminar)
- *Coupled ocean/sea ice circulation in the Antarctic Slope Current due to topographic eddy suppression and sea ice momentum redistribution*, Antarctic Margins Meeting at ANU, UNSW, and CSIRO, Mar 29, 2021. (Seminar)
- *What drives the ice-ocean system in the Antarctic Slope Current?* California Geophysical Fluid Dynamics (CalGFD) Meeting, Aug 20-21, 2020. (Talk)
- *Role of Tides in Ice-Ocean Interactions Over the Antarctic Continental Shelf and Slope*, AGU Ocean Sciences Meeting, San Diego, CA, Feb 16-21, 2020. (Poster)
- *Warming asynchronously rotating planets around low-mass stars by strong ocean tidal mixing*, the 5th Conference on Earth System Science, Shanghai, Jul 2-4, 2018. (Poster)
- *Tide, Ocean and Climate on Exoplanets*, AGU Fall Meeting, New Orleans, LA, Dec 11-15, 2017. (Poster)
- *Exploring the Dynamics Linking Arctic Melting to Eurasian Cold Winters from 1911 to 1940*, the 34th Chinese Meteorological Society Annual Meeting, Zhengzhou, Sep 26-29, 2017. (Talk)